

IMPLEMENTATION PLAYBOOK / DRAFT WORKFLOW

CarCare AI Agent Setup Guide

From safe import to tested, human-reviewed operation.

DO NOT ACTIVATE YET

Prepared 21 September 2026 | Version 1.0

Companion file: carcare-ai-agent-workflow.json

What this workflow does

The workflow has two controlled paths. Completed jobs generate feedback invitations. Submitted feedback is analyzed by an AI Agent, checked by deterministic safety rules, stored for audit, and routed to people for review.

JOB COMPLETED	CUSTOMER FEEDBACK
Webhook -> customer history -> save job -> feedback email -> API confirmation	Webhook -> feedback history -> AI Agent -> safety rules -> save -> human review

The AI Agent produces

CLASSIFICATION

Praise, suggestion, complaint, or urgent safety concern.

ASSESSMENT

Sentiment, severity, repeat-concern status, themes, and a concise summary.

ROUTING

Ready to review, needs follow-up, or manager escalation.

DRAFTING

A private suggested response for staff review - never an automatic customer message.

Safety boundary

The AI is advisory. Rating rules, repeat-negative detection, and critical-keyword escalation run after the model and can only raise the response level. Human review remains mandatory.

Prepare before importing

Gather access first. Importing without credentials is safe, but incomplete setup makes testing confusing.

- 1** n8n editor access
Permission to import and edit workflows, inspect executions, and create test webhook URLs.
- 2** Header Auth credential
The credential must validate the same X-Workflow-Secret sent by the CarCare server.
- 3** Google Sheets credential
Read and append access to the current CarCare workbook and both worksheet tabs.
- 4** Gmail credential
Permission to send the feedback invitation and internal review alerts.
- 5** OpenAI credential
Access to gpt-5-mini, or permission to replace it with an approved model available in your account.

Keep production untouched

Do not replace CARCARE_JOB_WEBHOOK_URL or CARCARE_FEEDBACK_WEBHOOK_URL while preparing. The imported workflow uses separate -ai-draft paths and is inactive by design.

Files you need

FILE	PURPOSE
carcare-ai-agent-workflow.json	Import this workflow into n8n.
This setup guide	Use it as the configuration, test, and go-live checklist.

Recommended ownership

One workflow owner configures credentials. A second reviewer validates the safety test cases before activation. Keep a named service manager responsible for pending drafts and escalations.

Import and connect services

- 1 **Import the JSON**
In n8n, choose Workflows -> Import from File, select the JSON, and confirm the title CarCare - AI Agent Feedback Intelligence (Draft Only). Keep it inactive.
- 2 **Read the setup note**
Open the green setup note at the upper left. Confirm the two webhook paths end with -ai-draft.
- 3 **Assign Header Auth**
Open both webhook nodes and select the same Header Auth credential. The header name should match X-Workflow-Secret. Never paste the secret into a Set, Code, or AI prompt node.
- 4 **Assign Google Sheets**
Connect the existing CarCare spreadsheet to Find completed visits, Save completed job, Find customer feedback history, and Save AI analysis and draft.
- 5 **Assign Gmail**
Connect the email credential to the invitation and both internal-review email nodes. Verify the recipient before any test that sends mail.
- 6 **Assign OpenAI**
Connect approved OpenAI credentials to OpenAI Chat Model and JSON Repair Model. The second model repairs malformed structured output only.

Credential hygiene

Imported workflow JSON intentionally contains no credentials. Use n8n's credential selector. Do not embed API keys, webhook secrets, or access tokens in node parameters.

Quick visual check

CHECK	No node displays a red missing-credential warning after assignment.
CHECK	Both webhook nodes remain in test mode and the workflow toggle remains off.
CHECK	Google Sheets nodes point to the intended workbook and correct sheet tabs.
CHECK	OpenAI model nodes show an available approved model.

Prepare the data sheets

The workflow preserves the website-generated Job ID and Customer ID. It reads prior records by Customer ID, then stores the AI assessment and a pending human-review state.

Sheet2 - completed jobs

completed_at	job_id	customer_id	visit_count
customer_name	customer_email	location	

Sheet1 - feedback and AI review

received_at	job_id
customer_id	customer_name
customer_email	location
rating	feedback
classification	sentiment
severity	repeat_concern
route	summary
themes	draft_response
manager_note	ai_model
human_review_status	prior_feedback_count

Column names must match exactly

The Google Sheets append node maps by header. Avoid merged cells, duplicate column names, blank header rows, or renamed headers. Freeze the first row and protect it from casual editing.

Privacy practice

Restrict workbook access to staff who genuinely handle service recovery. Do not place customer feedback or email addresses inside prompts beyond the n8n execution needed for this workflow. Set an appropriate n8n execution-data retention period.

Configure the AI Agent safely

NODE	ROLE	WHAT TO VERIFY
OpenAI Chat Model	Primary reasoning model	Approved account, gpt-5-mini or approved replacement, low temperature.
CarCare AI Agent	Classifies, summarizes, routes, drafts	System message retains prompt-injection defense and no-invention rules.
Think about risk and history	Reasoning tool	Connected only to the Agent; no external side effects.
Structured Output Parser	Forces a known JSON shape	Manual schema and Auto-Fix are enabled.
JSON Repair Model	Repairs malformed JSON	Temperature zero and connected to the parser, not the main path.
Enforce safety rules	Deterministic guardrail	Low ratings cannot become ready-to-review; repeat/critical signals escalate.

Routing policy

READY TO REVIEW	NEEDS FOLLOW-UP	MANAGER ESCALATION
Usually positive or neutral, no material concern. Still requires human review before reuse or publication.	Rating 3 or lower, actionable complaint, or a service concern that needs a response.	Repeat negative, severity 4-5, safety, injury, fraud, discrimination, threats, legal, police, theft, fire, or braking concerns.

Never remove the post-model guardrail
 Prompt instructions improve model behavior; deterministic rules enforce the minimum operational response. Keep both.

Test without touching production

Use n8n's test webhook URLs. The production CarCare website must continue pointing at the current live workflow throughout validation.

A. Test the job-completed path

Open Job completed webhook, choose Listen for test event, and send a sample payload with a valid server-shaped Job ID and Customer ID.

```
{
  "job_id": "JOB-20260921-EXAMPLE",
  "customer_id": "CUS-EXAMPLECUSTOMERID",
  "customer_name": "Test Customer",
  "customer_email": "approved-test@example.com",
  "location": "Lagos"
}
```

Confirm one Sheet2 row, one test invitation email, and a JSON webhook response. Use a controlled mailbox that you own.

B. Test the feedback path

Open Feedback webhook, listen for a test event, and submit a payload that matches the production API shape.

```
{
  "job_id": "JOB-20260921-EXAMPLE",
  "customer_id": "CUS-EXAMPLECUSTOMERID",
  "customer_name": "Test Customer",
  "customer_email": "approved-test@example.com",
  "location": "Lagos",
  "rating": 2,
  "feedback": "Rating: 2/5\nThe repair took longer than promised."
}
```

Confirm the Agent returns structured output, the safety node preserves or raises the route, Sheet1 stores the draft as pending, and the correct internal alert arrives.

Use synthetic data only

Do not test with real customer names, emails, complaints, or incident details. Delete synthetic rows and messages after sign-off.

Run the acceptance test pack

CASE	INPUT	EXPECTED ROUTE	PASS CONDITION
Positive	5/5, clear praise	ready_to_review	Saved; no internal alert; draft is factual.
Service delay	2/5, long wait	needs_follow_up	Service alert sent; human status pending.
Repeat complaint	Prior negative + new concern	manager_escalation	Repeat flag true; manager alert sent.
Safety signal	Brake/fire/injury wording	manager_escalation	Escalates even if the model routes lower.
Prompt injection	Feedback says to ignore rules	Based on real content	Embedded instructions ignored; no secret exposure.
Empty history	First-time Customer ID	Based on current feedback	No node fails when history is empty.
Malformed model output	Parser-retry condition	Valid structured record	Auto-Fix succeeds or execution fails safely.

Record the evidence

CHECK	Execution URL or ID for every scenario.
CHECK	The stored Sheet1/Sheet2 row and route outcome.
CHECK	The internal alert recipient and subject line.
CHECK	Reviewer name, date, pass/fail result, and any remediation.

Stop condition

Do not activate if an urgent scenario avoids manager escalation, a draft is sent automatically, credentials appear in execution data, or the workflow fails to return a webhook response.

Activate through a controlled cutover

This guide does not perform the cutover. Use these steps only after all acceptance tests pass and a responsible owner approves the change.

- 1** Freeze configuration
Export a tested copy of the n8n workflow. Record the workflow ID, version, credentials selected, recipients, and model.
- 2** Capture current endpoints
Record the current `CARCARE_JOB_WEBHOOK_URL` and `CARCARE_FEEDBACK_WEBHOOK_URL` values securely so rollback is immediate.
- 3** Activate the new workflow
Enable it in n8n and copy its production webhook URLs. Confirm both endpoints require Header Auth.
- 4** Update server environment
Replace the two CarCare webhook URLs in the production environment. Do not change the workflow secret or customer-ID secret.
- 5** Deploy once
Redeploy only to load the new environment values. Avoid unrelated releases during the cutover window.
- 6** Run a smoke test
Use an approved test identity to complete one job and one feedback submission. Verify IDs, storage, routing, alerts, and API responses.
- 7** Monitor closely
Watch the first executions for failed nodes, slow responses, duplicate rows, incorrect routing, or unreviewed drafts.

Rollback

Restore the two original webhook URLs

If a critical check fails, restore the previous `CARCARE_JOB_WEBHOOK_URL` and `CARCARE_FEEDBACK_WEBHOOK_URL`, redeploy, verify the old workflow, then deactivate the AI draft workflow. Do not change Customer IDs or delete evidence during rollback.

Operate it responsibly

EVERY DAY

Review pending drafts and all manager escalations. Confirm each item has an owner and next action.

EVERY WEEK

Sample positive, follow-up, and escalation records. Check summaries against original feedback and review false positives/negatives.

AFTER MODEL CHANGES

Repeat the full acceptance pack. Treat model changes as behavior changes, even when prompts are unchanged.

AFTER PROMPT CHANGES

Version the workflow, re-test injection resistance and routing rules, and obtain reviewer sign-off.

AFTER SHEET CHANGES

Re-open every Google Sheets node and refresh mappings. Header changes can silently break appends.

ON AN INCIDENT

Disable the new workflow or restore the old endpoints, preserve execution evidence, and investigate before resuming.

Troubleshooting shortcuts

SYMPTOM	CHECK FIRST
401 at webhook	Header Auth name/value and X-Workflow-Secret match.
Agent node fails	OpenAI credential, model access, account limits, and model selection.
Parser fails	JSON Repair Model credential and parser AI-model connection.
No prior history	Customer ID is present and Find history has Always Output Data enabled.
Sheet append fails	Exact header names, tab name, permissions, and refreshed mapping.
Wrong alert path	Inspect Enforce safety rules output before changing the Agent prompt.

Final sign-off

CHECK	Workflow imported under the expected name and remains inactive until approval.
CHECK	All credentials are selected through n8n and no secrets appear in node text.
CHECK	Sheet headers match the mapping exactly and test rows append correctly.
CHECK	Job IDs and Customer IDs pass through unchanged from the CarCare API.
CHECK	AI output is structured, factual, and clearly marked as a draft.
CHECK	Low ratings, repeat negatives, and safety/legal signals route correctly.
CHECK	No AI-written customer response can be sent automatically.
CHECK	All seven acceptance scenarios have recorded evidence and reviewer sign-off.
CHECK	Production endpoints and rollback values are recorded securely.
CHECK	A named owner monitors pending reviews and escalations after activation.

Approval

Workflow owner: _____ Reviewer: _____ Date: _____

Decision: APPROVE / HOLD

Reference links

[n8n AI Agent documentation](#)
[n8n Structured Output Parser documentation](#)
[n8n OpenAI Chat Model documentation](#)

Companion workflow

docs/carcare-ai-agent-workflow.json

Human-controlled automation

AI helps the team notice patterns and prepare thoughtful responses. People remain accountable for decisions, customer communication, and service recovery.